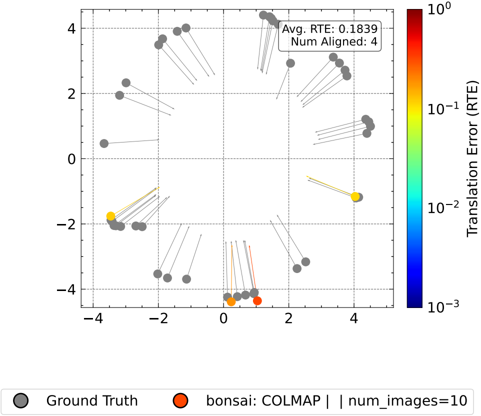
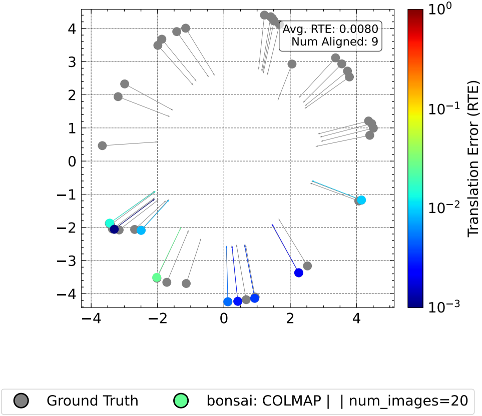
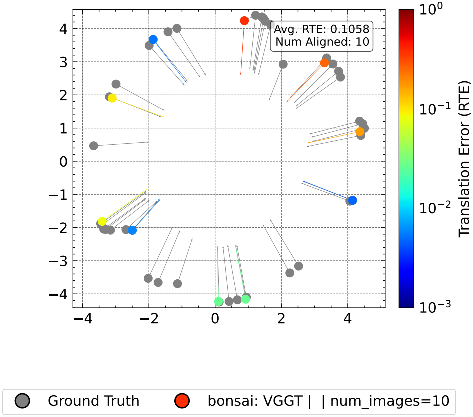
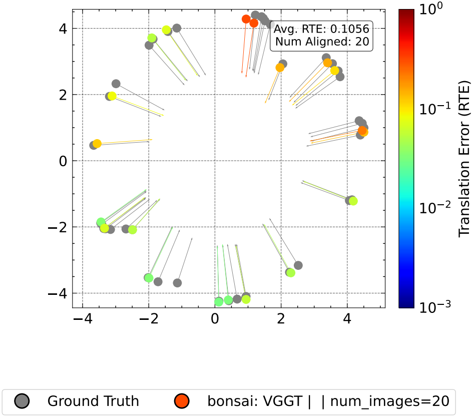
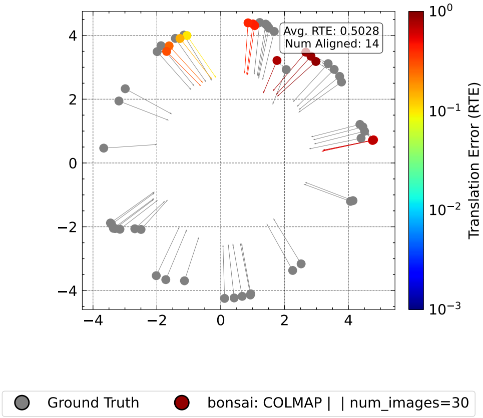
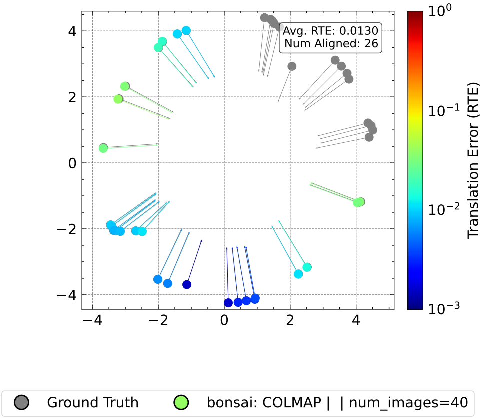

Avg. RTE: 0.1839
Num Aligned: 4
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
−4
−2
0
2
4
Ground Truth
bonsai: COLMAP | | num_images=10

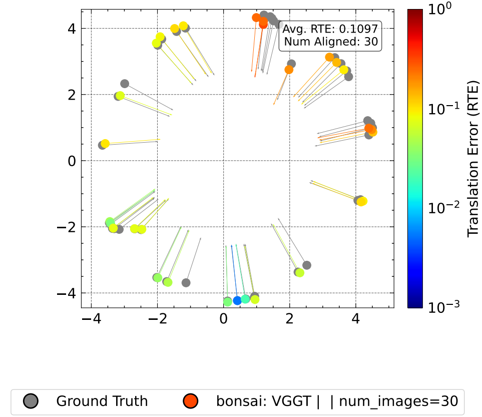

Avg. RTE: 0.0080
Num Aligned: 9
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
−4
−3
−2
−1
0
1
2
3
4
Ground Truth
bonsai: COLMAP | | num_images=20

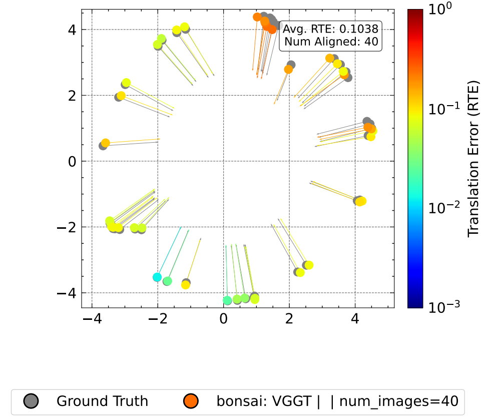

Avg. RTE: 0.1058
Num Aligned: 10
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
−4
−3
−2
−1
0
1
2
3
4
Ground Truth
bonsai: VGGT | | num_images=10

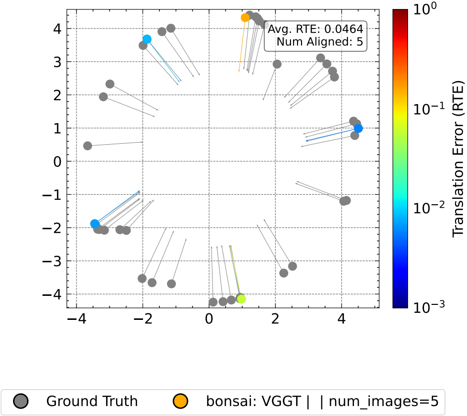

Avg. RTE: 0.1056
Num Aligned: 20
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
−4
−2
0
2
4
Ground Truth
bonsai: VGGT | | num_images=20

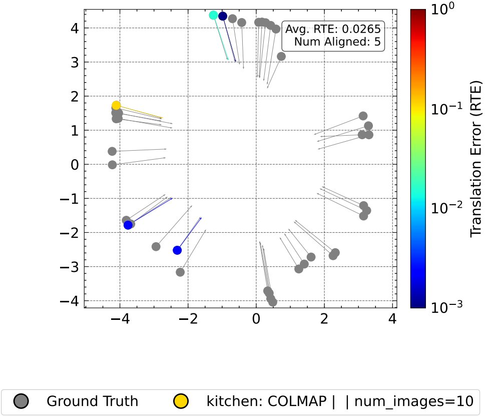

Avg. RTE: 0.5028
Num Aligned: 14
4
2
0
−2
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
bonsai: COLMAP | | num_images=30

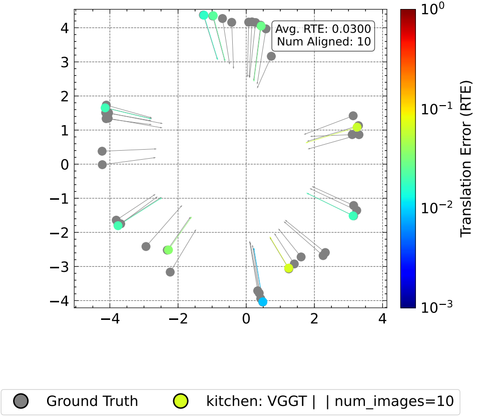

Avg. RTE: 0.0130
Num Aligned: 26
4
2
0
−2
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
bonsai: COLMAP | | num_images=40

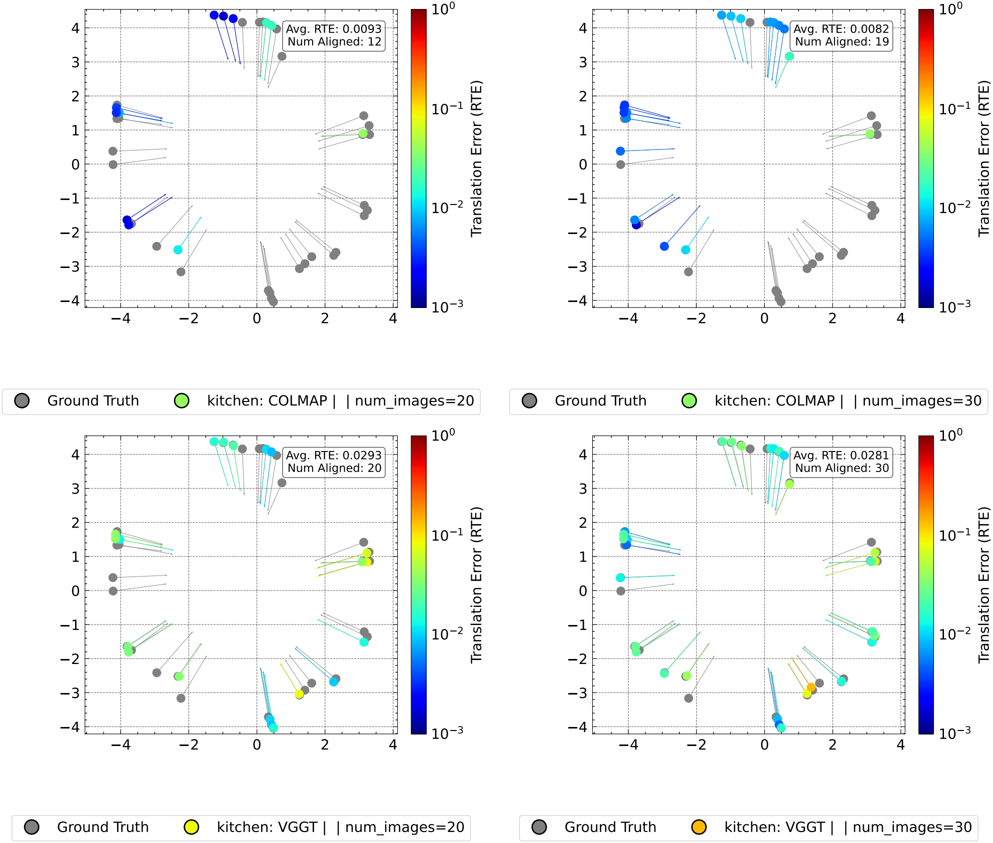

Avg. RTE: 0.1097
Num Aligned: 30
4
2
0
−2
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
bonsai: VGGT | | num_images=30

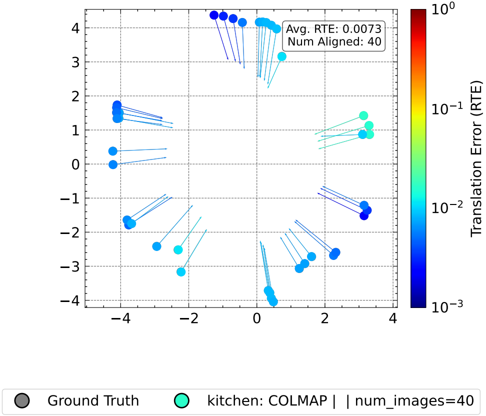

Avg. RTE: 0.1038
Num Aligned: 40
4
2
0
−2
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
bonsai: VGGT | | num_images=40

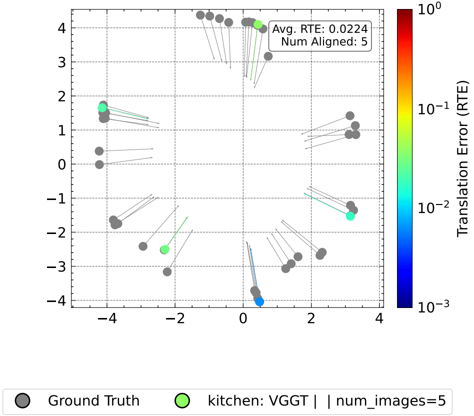

Avg. RTE: 0.0464
Num Aligned: 5
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
−4
−2
0
2
4
−4
−3
−2
−1
0
1
2
3
4
Ground Truth
bonsai: VGGT | | num_images=5

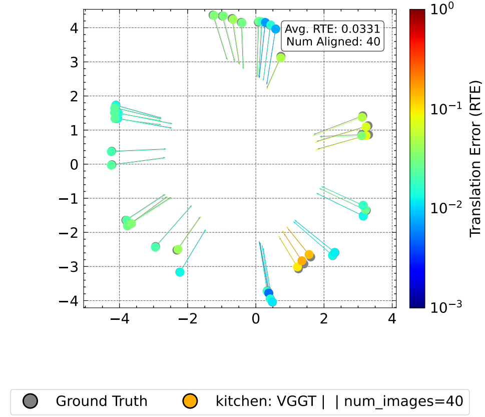

Avg. RTE: 0.0265
Num Aligned: 5
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
−4
−2
0
2
4
−4
−3
−2
−1
0
1
2
3
4
Ground Truth
kitchen: COLMAP | | num_images=10

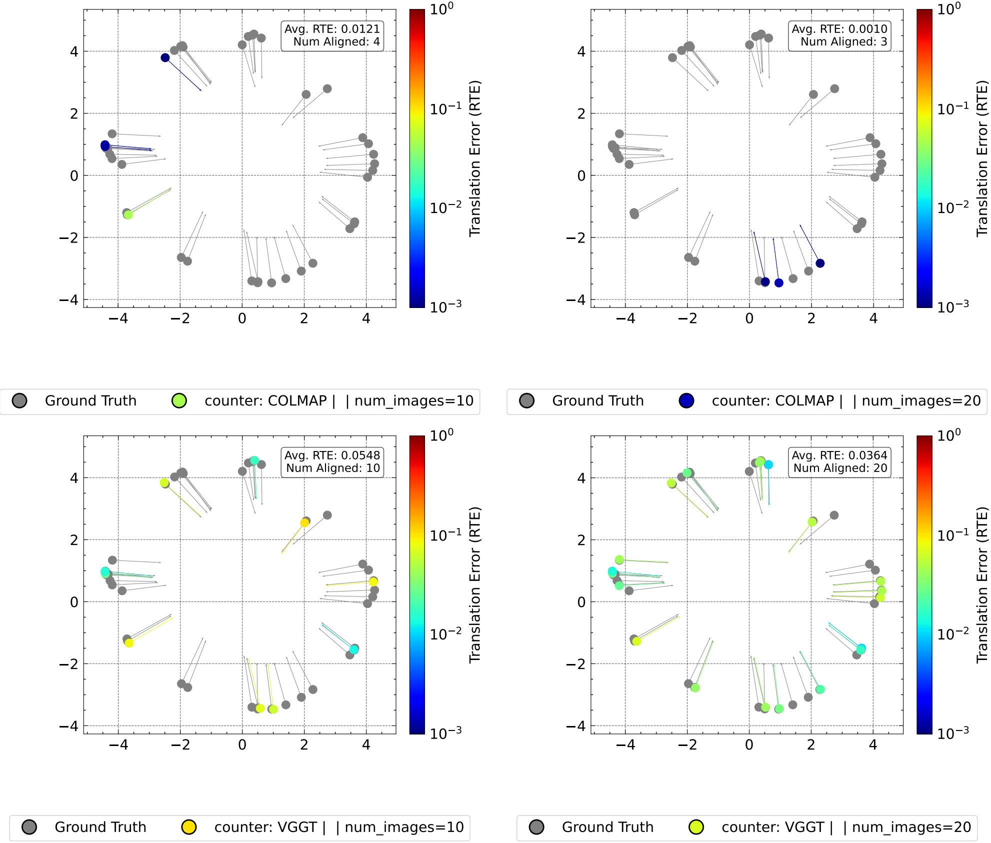

Avg. RTE: 0.0300
Num Aligned: 10
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
−4
−2
0
2
4
−4
−3
−2
−1
0
1
2
3
4
Ground Truth
kitchen: VGGT | | num_images=10

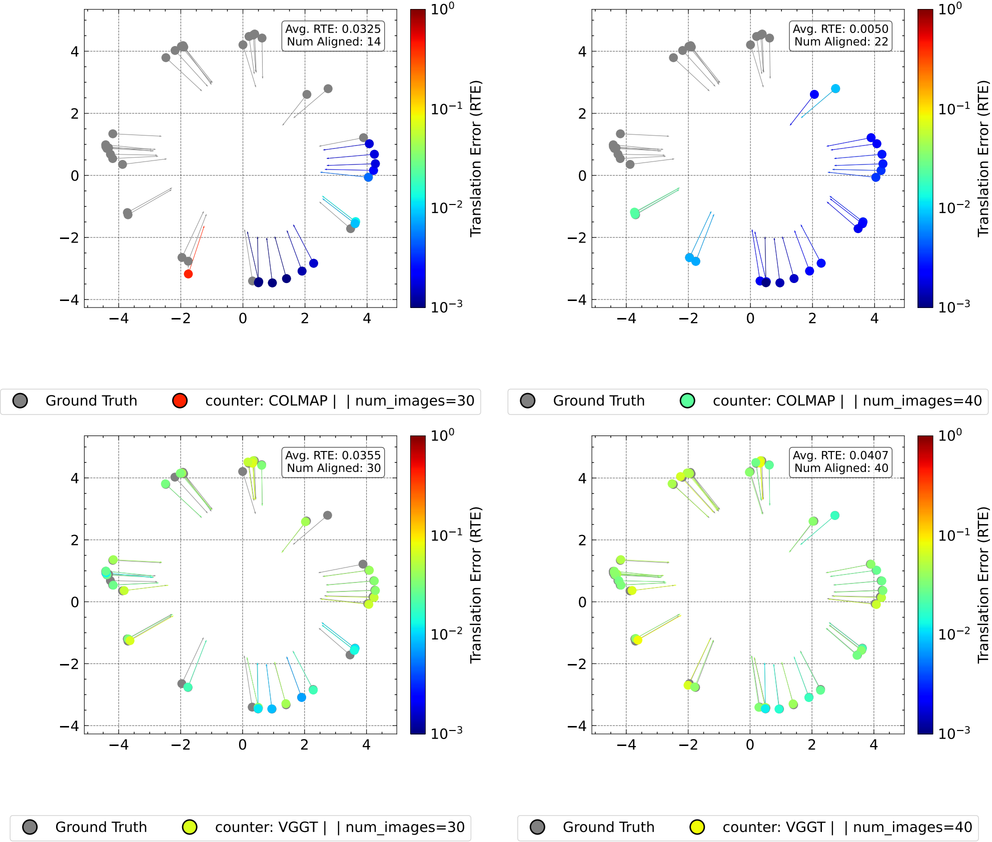

Avg. RTE: 0.0093
Num Aligned: 12
Translation Error (RTE)
10^0
10^−1
10^−2
10^−3
Ground Truth
kitchen: COLMAP | | num_images=20
Avg. RTE: 0.0082
Num Aligned: 19
Translation Error (RTE)
Ground Truth
kitchen: COLMAP | | num_images=30
Avg. RTE: 0.0293
Num Aligned: 20
Translation Error (RTE)
Ground Truth
kitchen: VGGT | | num_images=20
Avg. RTE: 0.0281
Num Aligned: 30
Translation Error (RTE)
Ground Truth
kitchen: VGGT | | num_images=30
−4
−2
0
2
4
−4
−3
−2
−1
0
1
2
3
4

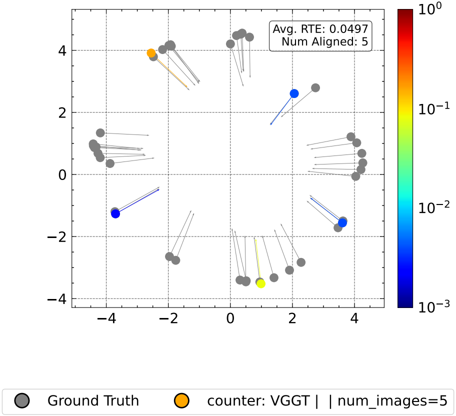

Avg. RTE: 0.0073
Num Aligned: 40
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
10^{0}
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
kitchen: COLMAP | | num_images=40

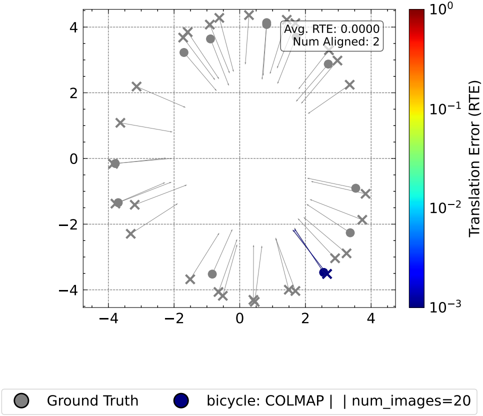

Avg. RTE: 0.0224
Num Aligned: 5
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
10^{0}
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
kitchen: VGGT | | num_images=5

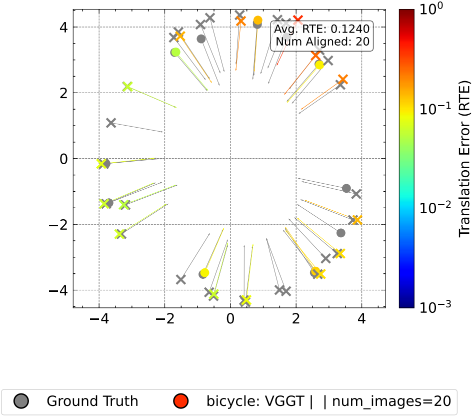

Avg. RTE: 0.0331
Num Aligned: 40
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
10^{0}
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
kitchen: VGGT | | num_images=40

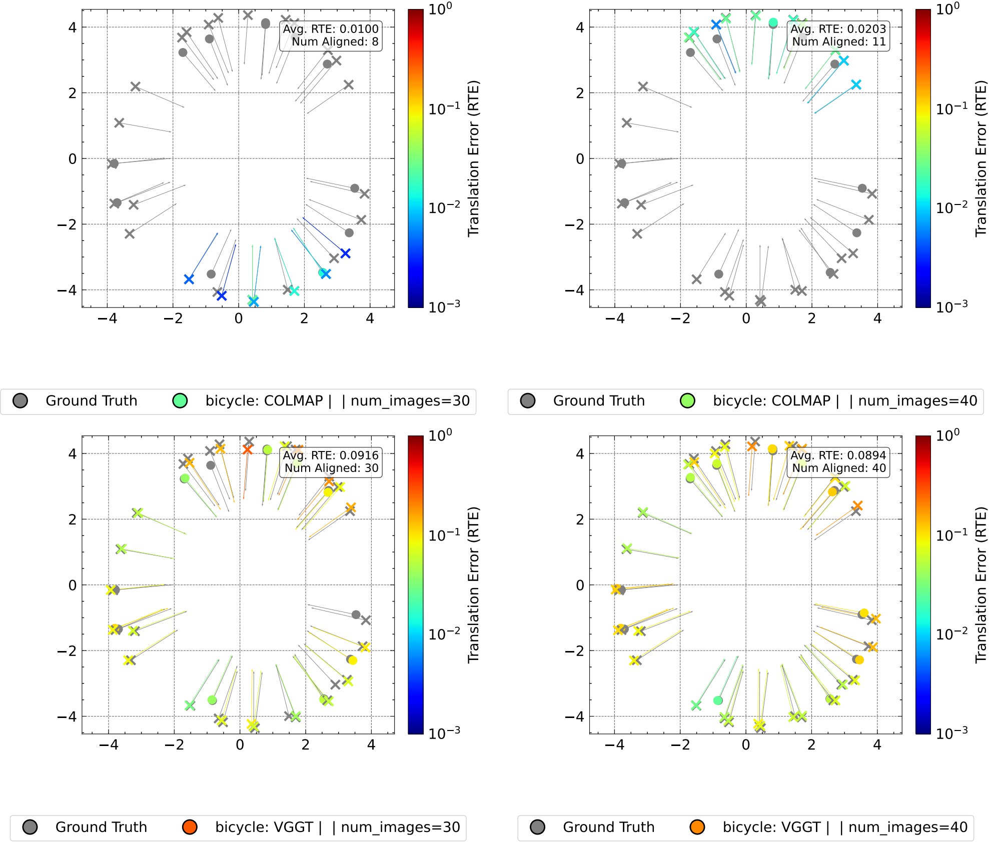

Avg. RTE: 0.0121
Num Aligned: 4
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
4
2
0
−2
−4
Ground Truth
counter: COLMAP | | num_images=10
Avg. RTE: 0.0010
Num Aligned: 3
Translation Error (RTE)
Ground Truth
counter: COLMAP | | num_images=20
Avg. RTE: 0.0548
Num Aligned: 10
Translation Error (RTE)
Ground Truth
counter: VGGT | | num_images=10
Avg. RTE: 0.0364
Num Aligned: 20
Translation Error (RTE)
Ground Truth
counter: VGGT | | num_images=20

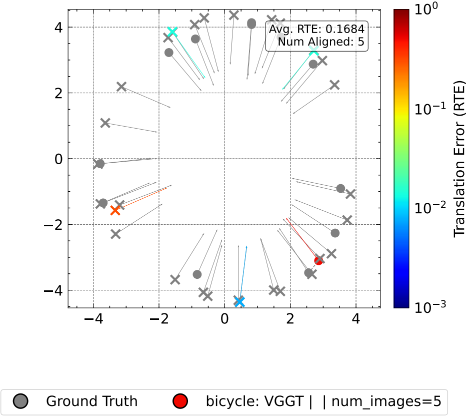

Avg. RTE: 0.0325
Num Aligned: 14
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
4
2
0
−2
−4
−4
−2
0
2
4
Ground Truth
counter: COLMAP | | num_images=30
Avg. RTE: 0.0050
Num Aligned: 22
Translation Error (RTE)
Ground Truth
counter: COLMAP | | num_images=40
Avg. RTE: 0.0355
Num Aligned: 30
Translation Error (RTE)
Ground Truth
counter: VGGT | | num_images=30
Avg. RTE: 0.0407
Num Aligned: 40
Translation Error (RTE)
Ground Truth
counter: VGGT | | num_images=40

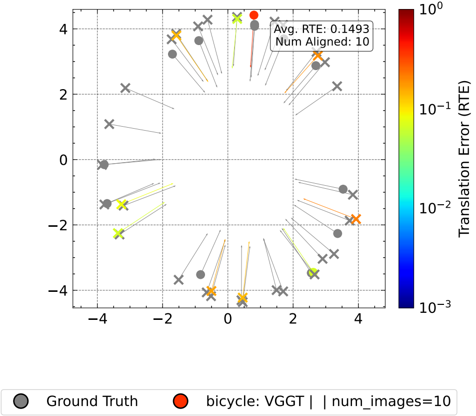

Avg. RTE: 0.0497
Num Aligned: 5
4
2
0
−2
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
counter: VGGT | | num_images=5

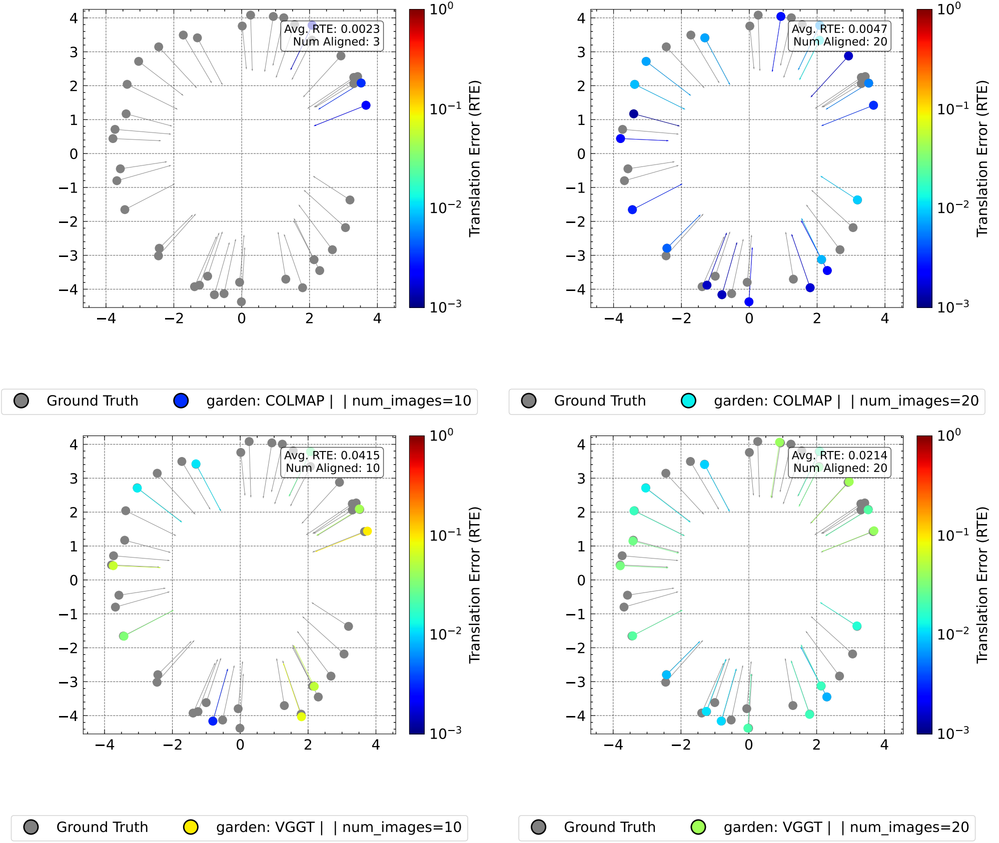

Avg. RTE: 0.0000
Num Aligned: 2
4
2
0
−2
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
bicycle: COLMAP | | num_images=20

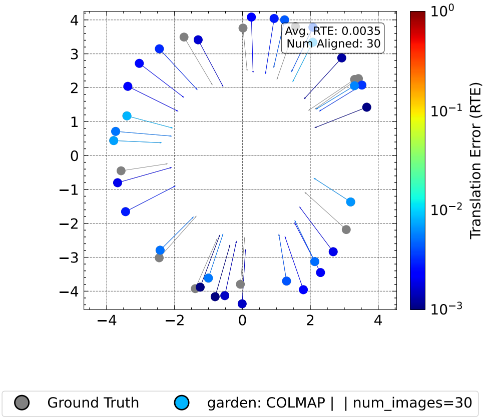

Avg. RTE: 0.1240
Num Aligned: 20
4
2
0
−2
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
bicycle: VGGT | | num_images=20

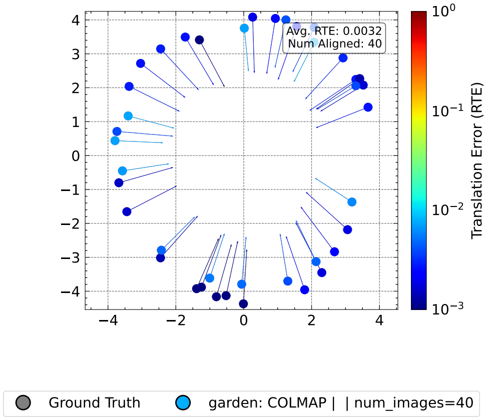

Avg. RTE: 0.0100
Num Aligned: 8
Translation Error (RTE)
Ground Truth
bicycle: COLMAP | | num_images=30
Avg. RTE: 0.0203
Num Aligned: 11
Translation Error (RTE)
Ground Truth
bicycle: COLMAP | | num_images=40
Avg. RTE: 0.0916
Num Aligned: 30
Translation Error (RTE)
Ground Truth
bicycle: VGGT | | num_images=30
Avg. RTE: 0.0894
Num Aligned: 40
Translation Error (RTE)
Ground Truth
bicycle: VGGT | | num_images=40
10^0
10^{-1}
10^{-2}
10^{-3}
−4
−2
0
2
4

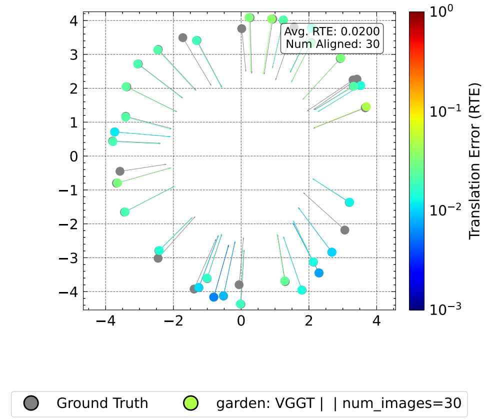

Avg. RTE: 0.1684
Num Aligned: 5
4
2
0
−2
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
bicycle: VGGT | | num_images=5

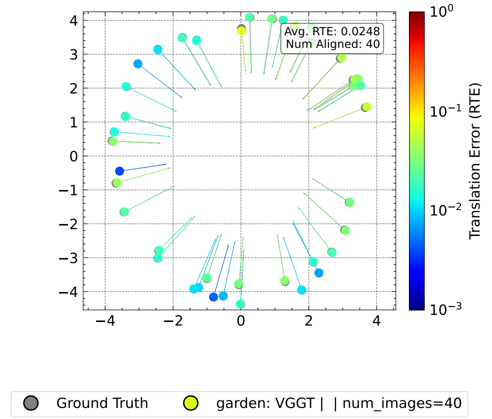

Avg. RTE: 0.1493
Num Aligned: 10
4
2
0
−2
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
bicycle: VGGT | | num_images=10

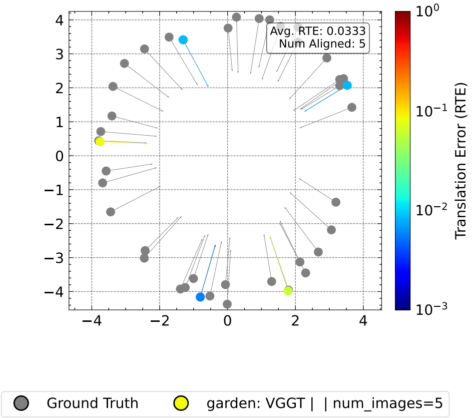

Avg. RTE: 0.0023
Num Aligned: 3
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
−4
−2
0
2
4
−4
−3
−2
−1
0
1
2
3
4
Ground Truth
garden: COLMAP | | num_images=10
Avg. RTE: 0.0047
Num Aligned: 20
Translation Error (RTE)
Ground Truth
garden: COLMAP | | num_images=20
Avg. RTE: 0.0415
Num Aligned: 10
Translation Error (RTE)
Ground Truth
garden: VGGT | | num_images=10
Avg. RTE: 0.0214
Num Aligned: 20
Translation Error (RTE)
Ground Truth
garden: VGGT | | num_images=20

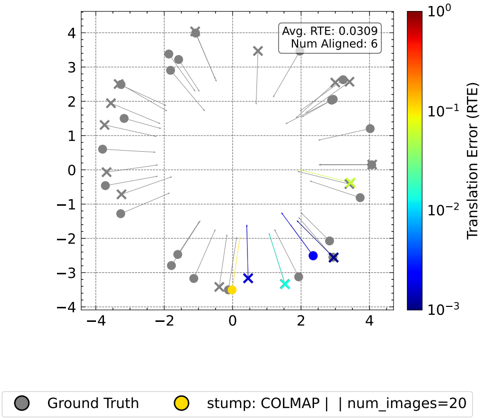

Avg. RTE: 0.0035
Num Aligned: 30
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
−4
−2
0
2
4
Ground Truth
garden: COLMAP | | num_images=30

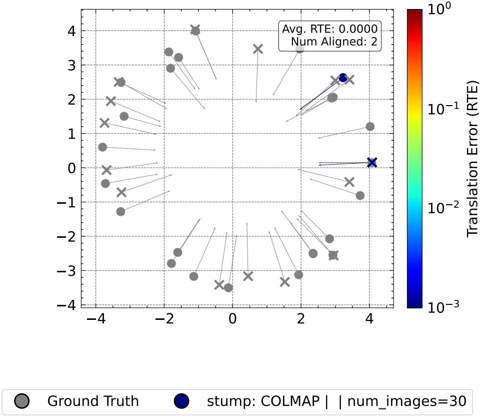

Avg. RTE: 0.0032
Num Aligned: 40
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
−4
−2
0
2
4
Ground Truth
garden: COLMAP | | num_images=40

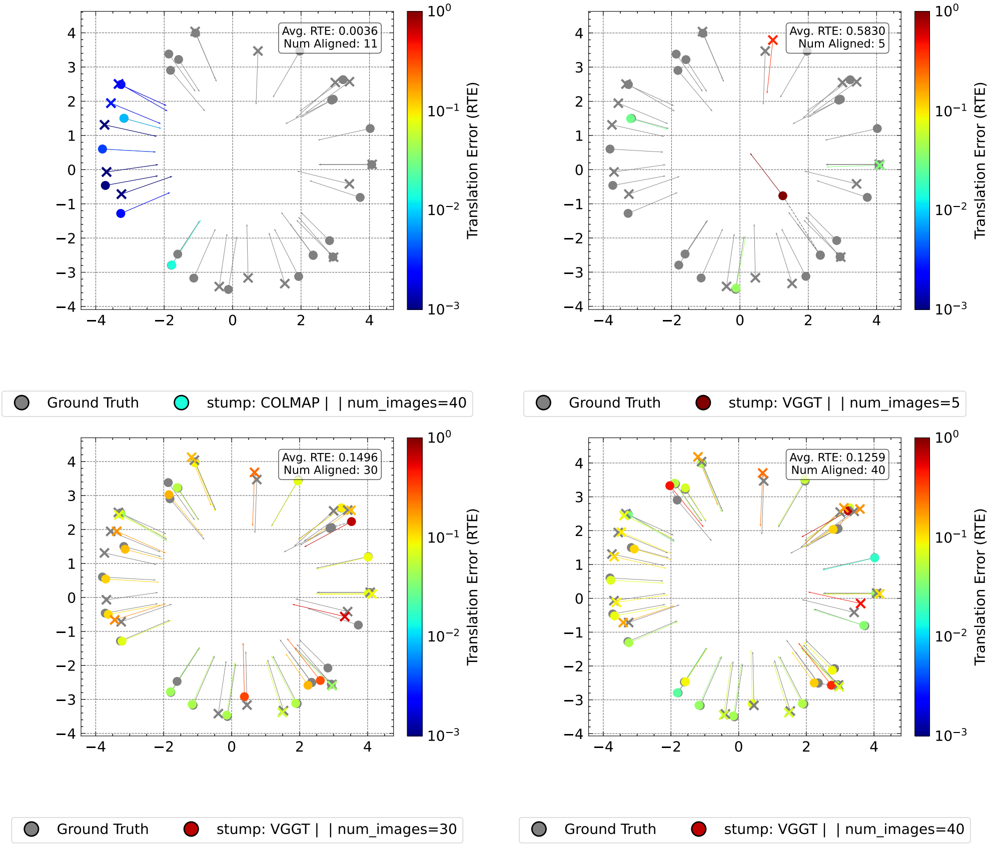

Avg. RTE: 0.0200
Num Aligned: 30
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
−4
−2
0
2
4
Ground Truth
garden: VGGT | | num_images=30

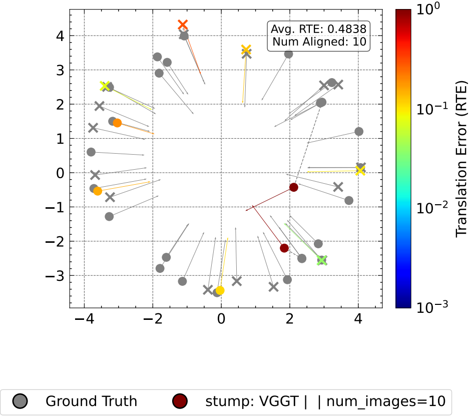

Avg. RTE: 0.0248
Num Aligned: 40
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
−4
−2
0
2
4
Ground Truth
garden: VGGT | | num_images=40

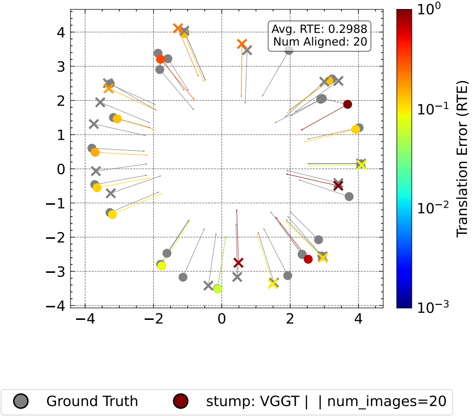

Avg. RTE: 0.0333
Num Aligned: 5
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
● Ground Truth
● garden: VGGT | | num_images=5

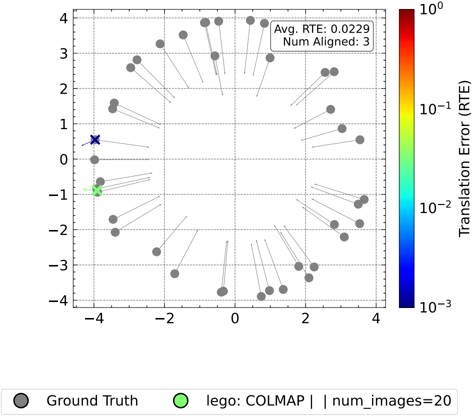

Avg. RTE: 0.0309
Num Aligned: 6
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
● Ground Truth
● stump: COLMAP | | num_images=20

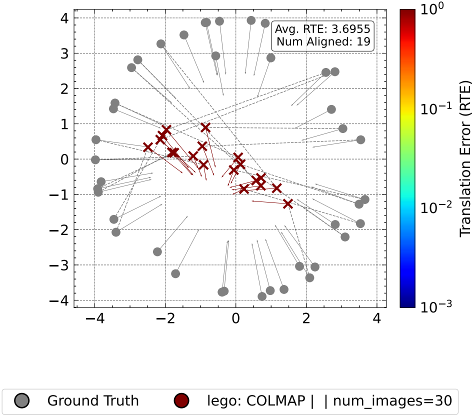

Avg. RTE: 0.0000
Num Aligned: 2
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
● Ground Truth
● stump: COLMAP | | num_images=30

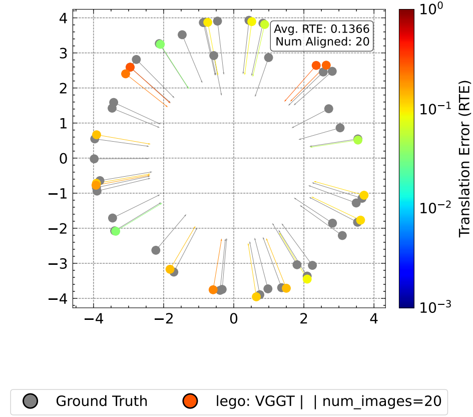

Avg. RTE: 0.0036
Num Aligned: 11
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
Ground Truth
stump: COLMAP | | num_images=40
Avg. RTE: 0.5830
Num Aligned: 5
Translation Error (RTE)
Ground Truth
stump: VGGT | | num_images=5
Avg. RTE: 0.1496
Num Aligned: 30
Translation Error (RTE)
Ground Truth
stump: VGGT | | num_images=30
Avg. RTE: 0.1259
Num Aligned: 40
Translation Error (RTE)
Ground Truth
stump: VGGT | | num_images=40
−4
−2
0
2
4
−4
−3
−2
−1
0
1
2
3
4

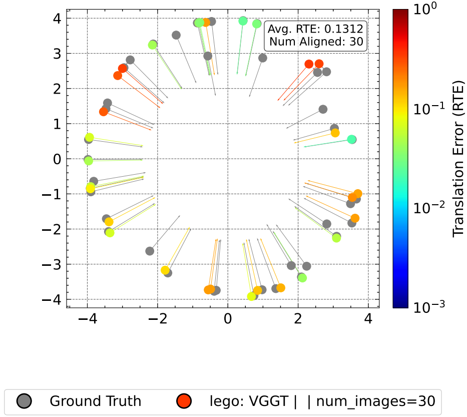

Avg. RTE: 0.4838
Num Aligned: 10
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
4
3
2
1
0
−1
−2
−3
−4
−2
0
2
4
Ground Truth
stump: VGGT | | num_images=10

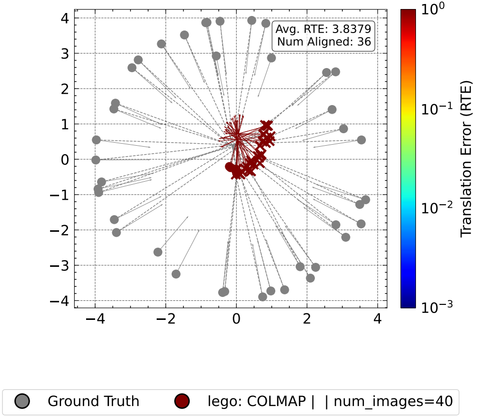

Avg. RTE: 0.2988
Num Aligned: 20
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
Ground Truth
stump: VGGT | | num_images=20

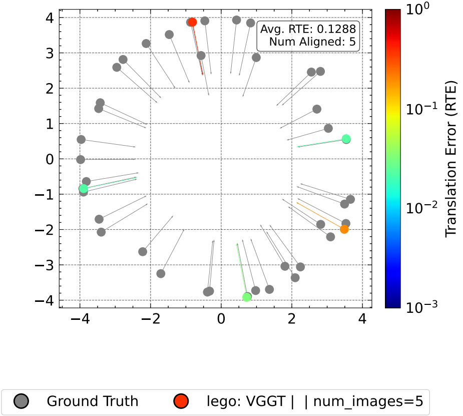

Avg. RTE: 0.0229
Num Aligned: 3
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
Ground Truth
lego: COLMAP | | num_images=20

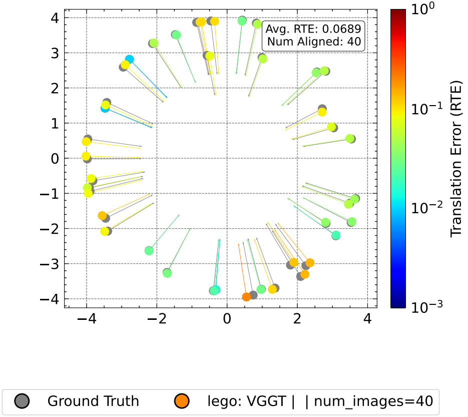

Avg. RTE: 3.6955
Num Aligned: 19
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
Ground Truth
lego: COLMAP | | num_images=30

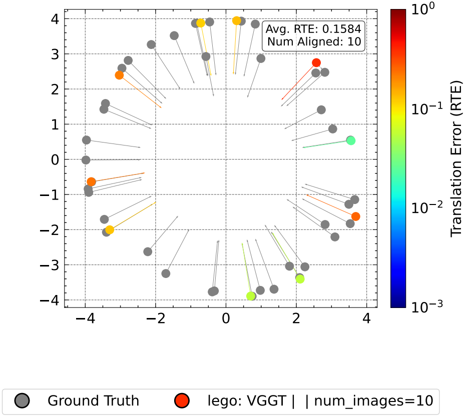

Avg. RTE: 0.1366
Num Aligned: 20
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
Ground Truth
lego: VGGT | | num_images=20

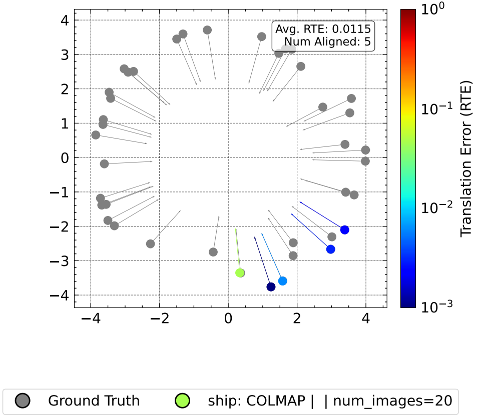

Avg. RTE: 0.1312
Num Aligned: 30
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
Ground Truth
lego: VGGT | | num_images=30

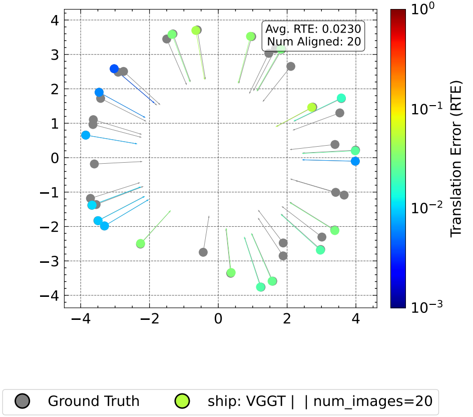

Avg. RTE: 3.8379
Num Aligned: 36
Translation Error (RTE)
10^0
10^-1
10^-2
10^-3
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
Ground Truth
lego: COLMAP | | num_images=40

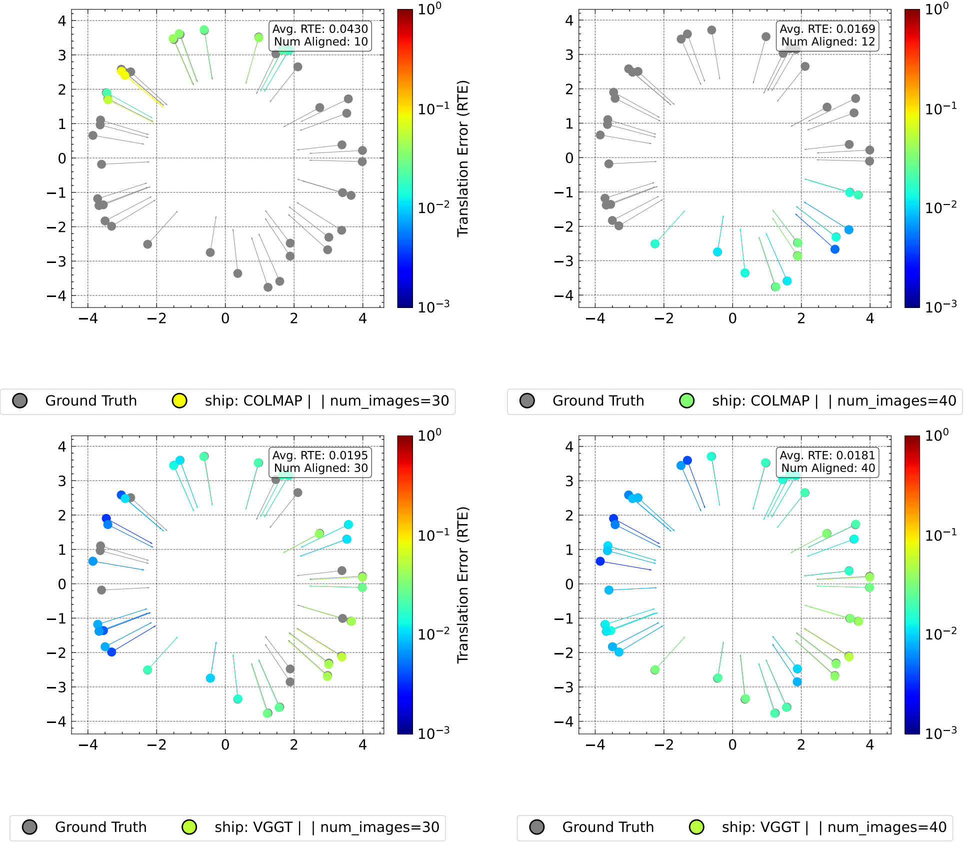

Avg. RTE: 0.1288
Num Aligned: 5
Translation Error (RTE)
10^0
10^-1
10^-2
10^-3
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
Ground Truth
lego: VGGT | | num_images=5

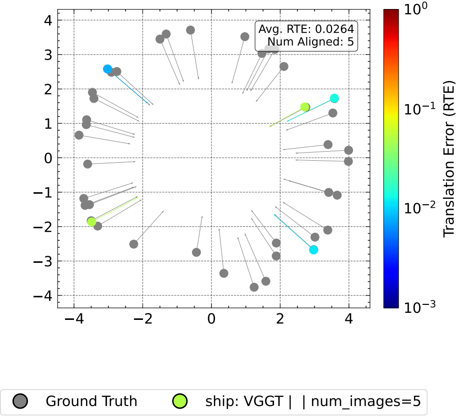

Avg. RTE: 0.0689
Num Aligned: 40
Translation Error (RTE)
10^0
10^-1
10^-2
10^-3
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
Ground Truth
lego: VGGT | | num_images=40

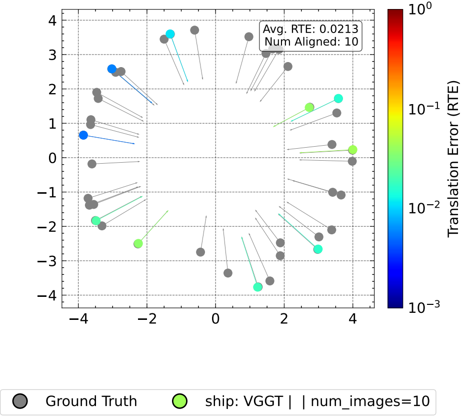

Avg. RTE: 0.1584
Num Aligned: 10
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
lego: VGGT | | num_images=10

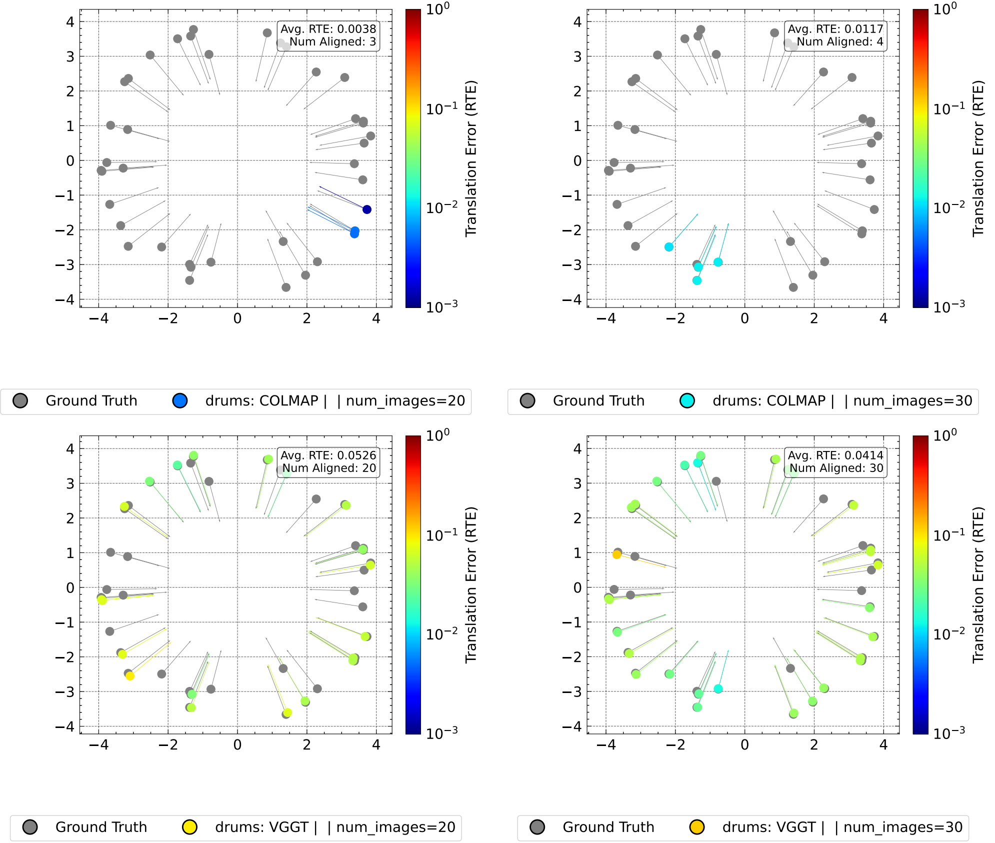

Avg. RTE: 0.0115
Num Aligned: 5
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
ship: COLMAP | | num_images=20

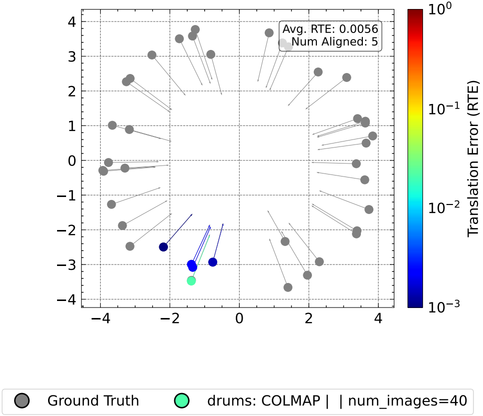

Avg. RTE: 0.0230
Num Aligned: 20
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
ship: VGGT | | num_images=20

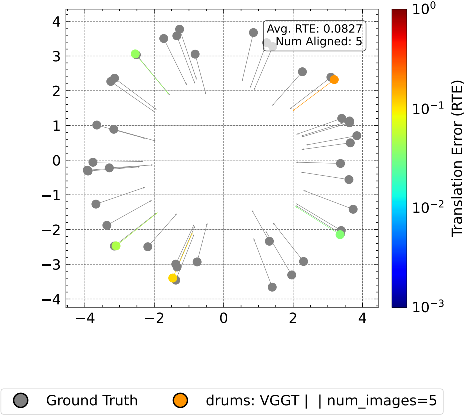

Avg. RTE: 0.0430
Num Aligned: 10
Translation Error (RTE)
Ground Truth
ship: COLMAP | | num_images=30
Avg. RTE: 0.0169
Num Aligned: 12
Translation Error (RTE)
Ground Truth
ship: COLMAP | | num_images=40
Avg. RTE: 0.0195
Num Aligned: 30
Translation Error (RTE)
Ground Truth
ship: VGGT | | num_images=30
Avg. RTE: 0.0181
Num Aligned: 40
Translation Error (RTE)
Ground Truth
ship: VGGT | | num_images=40
10^0
10^{-1}
10^{-2}
10^{-3}
−4
−3
−2
−1
0
1
2
3
4

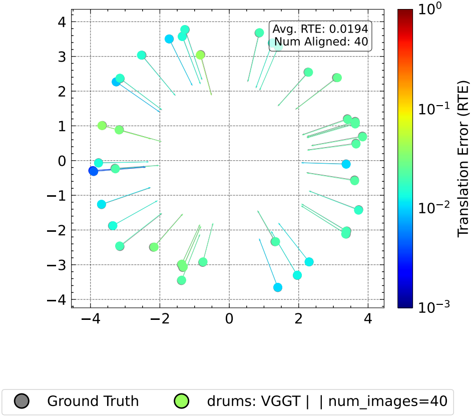

Avg. RTE: 0.0264
Num Aligned: 5
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
ship: VGGT | | num_images=5

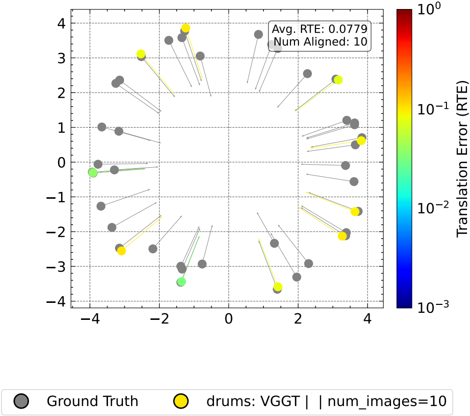

Avg. RTE: 0.0213
Num Aligned: 10
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
ship: VGGT | | num_images=10


Avg. RTE: 0.0038
Num Aligned: 3
Avg. RTE: 0.0117
Num Aligned: 4
Avg. RTE: 0.0526
Num Aligned: 20
Avg. RTE: 0.0414
Num Aligned: 30
Translation Error (RTE)
10^0
10^{-1}
10^{-2}
10^{-3}
−4
−2
0
2
4
● Ground Truth ● drums: COLMAP | | num_images=20
● Ground Truth ● drums: COLMAP | | num_images=30
● Ground Truth ● drums: VGGT | | num_images=20
● Ground Truth ● drums: VGGT | | num_images=30


Avg. RTE: 0.0056
Num Aligned: 5
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
drums: COLMAP | | num_images=40


Avg. RTE: 0.0827
Num Aligned: 5
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
drums: VGGT | | num_images=5


Avg. RTE: 0.0194
Num Aligned: 40
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
drums: VGGT | | num_images=40


Avg. RTE: 0.0779
Num Aligned: 10
4
3
2
1
0
−1
−2
−3
−4
−4
−2
0
2
4
10^0
10^{-1}
10^{-2}
10^{-3}
Translation Error (RTE)
Ground Truth
drums: VGGT | | num_images=10